

Open Source and Blockchain: Building Trust in a Decentralised World

The marriage of blockchain and open source has ushered in a new era of innovation, collaboration, and decentralised possibilities. So let's explore what blockchain entails, its diverse types, the dichotomy of public and private blockchains, and why open source is the bedrock upon which the future of blockchain will be built.



At its core, blockchain is a digital ledger that records transactions in a secure, transparent, and immutable manner. Picture it as a chain of blocks, each containing a set of transactions. These blocks are linked together in chronological order, forming an unbreakable chain. The brilliance of blockchain lies in its decentralised nature; no single entity has complete control over the network. Instead, it's maintained by a network of computers, or nodes, that work collaboratively to validate transactions through consensus mechanisms like proof of work (PoW) or proof of stake (PoS).

Let's take a practical example to understand this better. Imagine you're buying a piece of jewellery. With traditional methods, it's hard to verify if the gemstone in that necklace is really as rare as is claimed. Enter blockchain. In a blockchain-powered supply

chain, each step of the gemstone's journey, from the mine to the store, is recorded in the blockchain. So, when you scan a QR code on that necklace, you get a detailed history of where the gemstone was sourced, cut, and even its authenticity certificates. This level of transparency builds trust—something that was difficult to achieve before.

Overview of how blockchain works

Now, let's take a closer look at the fundamental mechanics of blockchain.

Blocks: Think of a blockchain as a chain of blocks, where each block is like a page in a ledger. It contains a list of transactions. For instance, in a cryptocurrency blockchain like Bitcoin, a block may record all the recent transactions that took place, including who sent how much Bitcoin to whom.

Nodes: Blockchain is not maintained by a central authority;

instead, it's a distributed network. This means there are many computers (nodes) connected to the network, each having a copy of the entire blockchain. These nodes work collectively to validate transactions and add them to the blockchain.

Consensus mechanism: The key to blockchain's security is its consensus mechanism. This is how the network ensures that all nodes agree on the validity of transactions. In Bitcoin, the most well-known consensus mechanism is proof of work (PoW). Miners, the participants in the network, solve complex mathematical puzzles to validate transactions. Once a puzzle is solved, the new block is added to the chain, and the miner is rewarded with some cryptocurrency. This process requires significant computational power, making it extremely difficult for anyone to manipulate the blockchain.